MECH MAINT- BOILER Department
SIMHADRI

Note No :1

Ref:SMTPP/MECH MAINT-/2023-24/LMI/843466 **Date:**23/01/2024(09:46:12)

Subject: Revision of APH LMI as per changes in OGN

The LMI on "Operation and Maintenance Practices of Air Preheaters " LMI/OPS/MECH/055 has been revised as per revised OGN "COS-ISO-00-OGN/OPS/MECH/055" Rev. No.: 1 Dec'2023.

Revised LMI and OGN is attached.

Put up for kind approval please.

Submitted by:

कृष्ण लाल - वरिष्ठ प्रबंधक

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Note No:2

Revisions incorporated as per revised COS-ISO-00-OGN/OPS/MECH/055 Rev. No.: 1 December 2023.

May plz approve.

Submitted by:

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Note No:3

Changes as per revised OGN "COS-ISO-00-OGN/OPS/MECH/055" have been incorporated in the revised ver.1 of LMI "LMI/OPS/MECH/055".

HOP may kindly accord approval for this revised version of subject LMI.

Submitted by:

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Note No:8

Please confirm whether all points as mentioned in the LMI, during overhauling of Unit-1 (recently concluded) have been taken care of, especially clause no. 5.5.10.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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Note No:10

May plz confirm as per note-8 of GM(Maint).

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Note No:11

Work has been completed as per protocol and LMI including clause no 5.5.10.

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Note No:12

May plz be approved.

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Note No:14

May please be approved

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Note No:15

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LMI/OPS/MECH/055

Rev. No. : 1 January 2024

LOCATION MANAGEMENT INSTRUCTION

Operation and Maintenance Practices of Air Preheaters

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER PROJECT

NTPC LIMITED

SIMHADRI SUPER THERMAL POWER PROJECT

TECHNICAL COMPLIANCE SYSTEM

LOCATION MANAGEMENT INSTRUCTION - LMI//OPS/MECH/055

Rev.No.:1 Date: January 2024

Operation and Maintenance Practices of Air Preheaters

Approved for
Implementation by ------------------------------
HOP (SSTPP)

Date:

Enquiries to : GM(Operation & Maintenance)

LMI APPROVAL DOCUMENT

1	TITLE OF LMI	OPERATION AND MAINTENANCE PRACTICES OF AIR PREHEATERS
2	DOCUMENT NUMBER REVISION/ ISSUE NO & DATE	LMI/OPS/MECH/055 Revision -1, January 2024
3	ORIGINAL/ REF DOCUMENT	COS-ISO-00-OGN/OPS/MECH/055 Rev. No.: 1 December 2023
4	PREPARED BY	SIGNATURE: ………………………. Krishan Lal, Sr. Mgr(BMD)
5	CHECKED BY	SIGNATURE: ………………………. Ritesh Kumar, Sr Mgr (BMD)
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	REVIEWED BY	SIGNATURE: ………………………. Dhurjati Prasad Patra, GM (O&M)
8	APPROVED FOR ISSUE	SIGNATURE: ………………………. Sanjay Kumar Sinha, HOP(SSTPP)
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Operation and Maintenance Practices of Air Preheaters

INDEX

Operation and Maintenance Practices of Air Preheaters

1.0 INTRODUCTION

This document aims at providing adequate guidelines for safe operation and maintenance practices for Air Preheaters. Some best practices for Air Preheaters have been incorporated in this document. With the implementation of the operation and maintenance practices as given in this document stations are bound to have better Air Preheater performance and substantial gain in boiler heat rate and efficiency.

2.0 SUPERSEDED DOCUMENTS

LMI/OPS/MECH/055, Revision -0, January 2022

3.0 REVISION DETAILS

a) Clause 5.1.5 : Added for replacement of sector plates with baskets
b) Clause 5.1.6 : Added for Checking of sector plates locking
c) Clause 5.1.19 : Added for CE Basket doors identification & tightening
d) Clause 5.2.19 : Added for Runner plate & support bearing size measurement
e) Clause 5.3.16 : Added related to Stitch welding on bracing protection angles
f) Clause 5.4.9 : Basket drying after hot washing modified.
g) Clause 5.5.10 : Added for checking of soot blower lance thickness

4.0 BEST PRACTICES

The following best maintenance practices are to be adopted for regenerative air preheaters. OEM recommendation for preventive / overhauling maintenance to be followed. Overhauling protocols / critical dimension are to be recorded / maintained as per OEM recommendations.

4.1 ACTIONS TO BE TAKEN FOR FREE THERMAL EXPANSION

- Rigorous inspection of hangers & supports.
- Removal of all restrictions after work from areas of possible expansion restriction.
- Expansion study with help of aluminum finger tabs.
- Applications of special purpose anti seize compound in specific areas.
- Proper rain guard cover on guide bearing area & support bearing area to avoid Water ingress.
- Healthiness of ash evacuation system of Economizer and APH hoppers should be ensured to improve the performance of APHs and ESP. If required, discharge pipe may be re-routed to minimize/avoid bends.

4.2. POSSIBLE AREAS TO BE INSPECTED TO AVOID THERMAL EXPANSION RESTRICTION

- Static seal
- Guide Bearing spool piece
- Air seal housing
- Sector plate
- Main drive rack pinion drive
- Guide bearing sleeve
- Rotor post seal

5. MAINTENANCE IN AIR PREHEATER

5.1. CORRECT METHOD OF RESTRICTING INTER SECTOR LEAKAGE:

5.1.1 Rotor leveling within 0.1 mm / mtr as recommended by OEM.

5.1.2 Sector plate replacement/repair.

5.1.3 PAPH sector plate to be replaced in case of any hole in sector plate and or plate grooves /undulations are more than 3mm.

5.1.4 SAPH only small holes at outboard/inboard may be repaired, otherwise holes in other portions and or grooves /undulations are more than 3mm it is to be replaced.

5.1.5 Basket replacement must be done as per criteria mentioned. All sector plates also to be replaced at the time of 100% basket replacement. Station may plan this activity in two APHs and then in others two APHs during next overhaul. If more erosion observed, then these may be replaced earlier also.

5.1.6 Sector Plates locking must be checked in every OH and to be done as per design after replacement of sector plates.

5.1.7 Sector plate leveling, within +/- 0.5mm.

5.1.8 T-bar leveling & centering within +/- 3 mm.

5.1.9 Seal setting as per manual.

5.1.10 Zero leakage through rotor diaphragm Worn out rotor diaphragms to be repaired by grafting new plates & grinding.

5.1.11 Cold end grills and hot-end supports are to be properly inspected and corrected.

5.1.12 Any gap between basket and diaphragm to be either covered by basket with seals or MS Strip.

5.1.13 Basket to Basket gap if more than 3mm to be covered by MS Strip.

5.1.14 T bar to be replaced if T bar groove is more than 1mm.

5.1.15 Proper alignment and run-out of T-bar to be checked and maintained within limit as per manual. Stitch weld may also be done after bolt tightening.

5.1.16 Trimming and welding of sealing tab so that by-pass should not get struck within T-bar.

5.1.17 Only HT bolts to be used in Air preheater like for seals, T-bar, CE basket covers etc.

5.1.18 One set of sealing surfaces should always be made available at site.

5.1.19 Marking of CE basket doors to be ensure for proper fitment, while removing. Also all studs of basket doors should be available for tightening with washers.

5.2 MAINTENANCE OF MAIN BEARINGS

5.2.1 Proper levelling within 0.1 mm / mtr as recommended by OEM.

5.2.2 Monthly oil monitoring for moisture & MI at station level and Quarterly oil monitoring through wear debris analysis at NETRA/ site chemistry. In case of any abnormality in the WDA (PQ Index>25), oil should be replaced immediately and fresh sample should be collected after 15-20days of operation and sent for analysis. If abnormality in the WDA persists then bearing inspection/replacement should be planned at the earliest/opportunity.

5.2.3 Oil sample to be collected with APH in running condition, when the oil is warm and under circulation.

5.2.4 One oil sample should be sent for Wear debris analysis before every overhaul.

5.2.5 If pre Overhaul PQ index is high(>25) , bearings(Guide/Support) to be inspected by removing from the position. Replacement to be done as per condition.

5.2.6 Cooler and filter inspection. Hydraulic testing of oil coolers should be done during every overhaul.

5.2.7 Controlling hot air leakage from air seal housing.

5.2.8 Protective water shield at support bearing. Ingress of ash and water into the bearing assembly should be arrested.

5.2.9 Oil flow view glass or flow meter to be provided wherever not available.

5.2.10 Fibroscopic /Borescope inspection of support bearing assembly may be done duringoverhauling.

5.2.11 Inspection of guide bearing should be done during every overhaul.

5.2.12 Torque tightness of locking plate holding bolts should be ensured. These bolts should be inspected during every overhaul.

5.2.13 During inspection of Guide bearings all associated components healthiness to be checked, specially oil ring/seal tube, if required it is to be replaced.

5.2.14 It should be ensured that linear expansion of rotor post is not arrested during maintenance work. Any restriction in expansion would put additional stress on the guide bearing.

5.2.15 Inspection of Support bearing & Guide bearing by withdrawal of bearing from its position after 10-12 years of operation.

5.2.16 Subsequent inspection of Support Bearing & Guide bearing by withdrawal of bearing from its position after another 4-6 years.

5.2.17 Further, the inspection and replacement cycle as described above shall be repeated for new replaced bearings.

5.2.18 Replacement of Support Bearings & Guide Bearings after 20 years irrespective of condition/WDA or make.

5.2.19 While replacing support bearing, Runner plate seating area OD & Support bearing top race ID must be measured and interference fit to be maintained as per OEM.

5.2.20 Induction coils to be used for heating the required components, during Support bearing replacement. It can save time significantly.

5.2.21 Replacement of bearings shall be with source-standardized bearings.

5.2.22 All EPK make bearings to be replaced irrespective of the condition.

5.2.23 Sampling location of lube oil for WDA/monitoring shall be done with proper arrangement of sampling tapping with valve as near to oil sump/ bearing housing / gear-box etc from the drain point.

5.2.24 Replacement of lub oil during every overhaul after flushing.

5.2.25 While replacing lube oil, viscosity of fresh oil should be checked to ensure that it is as per specification.

5.2.26 When replacing the lubricant in the rotor bearing, the oil level should be checked after the lubricant has been able to heat up and circulate through the system. Failure to do this may result in a false oil level reading, which could lead to the bearing being insufficiently lubricated during service.

5.2.27 In-situ inspection should be done by NDT (DPT/UT/MPI) of Guide trunnion near oil deflector plate. Oil deflector plate should be fixed with screws on the sleeve and not to be welded on guide trunnion.

5.2.28 Torque tightness of Guide bearing locking plate holding bolts should be ensured. These bolts should be inspected during every overhaul.

5.2.29 When doing welding on the air preheater, care should be taken to ensure the ground is made from the rotor structure to the rotor frame and not grounded through the support bearings or radial guide bearings.

5.2.30 In PAPH guide bearing area some reliable sealing applications may be used like AIR Seal etc.

Note: If WDA facility is available at site chemistry, and quarterly schedule is followed, Limits for PQ Index to be followed as per NETRA standards.

5.3 MAINTENANCE OF DUCTS & DAMPERS

5.3.1 Insulation survey of FG and hot air ducts shall be conducted before OH and recommendations of the survey should be implemented.

5.3.2 Operation of dampers may be checked at the time of unit shut down. APH gas inlet dampers to be operated to remove accumulated ash, if any.

5.3.3 Proper scaffolding to be made for inspection of ducts at inlet and outlet of APH for erosion. Physical inspection of 100% bracings for erosion to be done.

5.3.4 Replacement of eroded duct, bracing and supports. Erosion resistant material and erosion prevention measures like coatings to be done in erosion prone area.

5.3.5 Damaged/eroded expansion bellow and its cover plate to be replaced in at least 1mtr length.

5.3.6 At APH inlet patching of plates should not be done. All eroded plates should be removed and new plates should be grafted and welded from all the sides to avoid falling of plates during running of APH.

5.3.7 The entire length of all the damaged / eroded bracing and supports to be replaced and welded fully to avoid falling of supports. No joints should be allowed.

5.3.8 All eroded guide vanes are to be replaced. Patching repair should not be done. Wire reinforced plastic refractory/erosion prevention coating at bottom portion and Tubes at guide vane inlet to be used.

5.3.9 Repair/replacement of all dampers and the sealing of these dampers. Flaps and shafts of inlet damper to be repaired / replaced in full if it is eroded. Patch plate on flap is not permitted, it should be repaired by grafting only.

5.3.10 Isolation of APHs must be ensured in all stations to avoid unit outage in the event of any breakdown of any APH. Eroded damper flaps should be replaced/ revived.

5.3.11 Multiple patches in FG duct should be removed. Damaged portions should be repaired by grafting using Sail hard plate.

5.3.12 Sail hard and wear resist plate to be used in erosion prone locations in duct. High erosion prone locations are to be further covered by wire reinforced plastic refractory with proper curing.

5.3.13 Welding of bracings/ diverter/ cover plate to be done checked and re-welding to be carried out on old weld even if bracings/ diverter/ cover plate are found OK.

5.3.14 Inspection of gusset plates at both ends of bracings for erosion & replacement of eroded members. Welding of gusset plate with FG/air duct and welding of gusset plate with bracing should be full (stitch welding should not be done).

5.3.15 10% DP test of welding of bracing gusset plates to be carried out.

5.3.16 Adequacy of stitch welding (3" long stitch x 12" pitch) of protection angle with bracing in staggered manner on both sides to be done.

5.3.17 All the redundant supports/channels/angles welded during repair work should be removed before giving clearance for trial run of APH.

5.3.18 After the completion of works, a joint inspection of FG duct, bracings and expansion joint protection plate should be done by BMD, FQA and Operation. A joint protocol to this effect should be made.

5.3.19 Any angle used as sacrificial member for erosion prevention should be of such size that its welding with member is not falling in gas flow path.

5.3.20 Station must prepare a rolling plan for replacement of Expansion joints (MEJs and NMEJs) in a time bound manner. MEJs should be replaced once in every 10 years or earlier, if damaged and NMEJs should be replaced in 6 years or earlier, if damaged.

5.3.21 Non-metallic expansion joints in cold and hot PA duct & hot Secondary air duct, wherever provided, should be replaced with Metallic expansion joints in phased manner. Stations must prepare a rolling plan accordingly. While converting NMEJs to MEN, the corrugations should not protrude in flow path.

5.3.22 CFD modeling of FG ducts to be planned as per requirement and necessary action on the basis of CFD study should be taken during overhaul.

5.3.23 During every opportunity, APHs internal inspection may be carried out for any foreign (Bracing/angle etc) material.

5.3.24 Ducts above and below APH up-to damper to be under APH maintenance group.

5.4 APH WASHING

5.4.1 Water washing of air heaters should be carried out in hot condition after boiler shut down i.e. when flue gas temperature is between 200 to 150 deg.C with draft fans in stopped condition. Coverage of water washing in all the heating elements is to be ensured before completion of water washing. pH value of outlet water may be measured to determine effectiveness of cleaning and to be compared with inlet water pH value.

5.4.2 APH washing to be done with APH wash water. During APH washing all fans to be kept stopped.

5.4.3 Before start of APH washing all drains in air side ducts to be opened and any chocking to be cleared.

5.4.4 APH air side ducts drains (size and numbers) are to be as per **Annexure-I**

5.4.5 Throughout the washing and drying APH should not be stopped.

5.4.6 Before start of APH washing, APH hopper manholes and its gate (KGV) to be opened and any chocking to be cleared.

5.4.7 No chocking to be ensured from Duct drains and APH hoppers to be ensured throughout the APH washing.

5.4.8 Hot washing of APH baskets should be carried out during available opportunity if DP across APH is more than 50% than design. Drying of baskets should be ensured before boiler light-up. Improvement in DPs pre-post of APH washing to be monitored.

5.4.9 Allow the rotor and heating element surfaces to dry completely, before returning the gas air heater into service. The recommended minimum drying period is 12 hrs without fans and 8 hrs with fans.

5.4.10 To shorten the required drying duration, the operator may choose to use one or more of the following methods.

a. Open all inlet and outlet dampers and access openings and allow the air heater to dry using natural air draft.

b. If dry compressed air can be made temporarily available to the soot blower, employ air blowing while rotating the rotor at reduced speed.

5.5 BASKETS MAINTENANCE

5.5.1 During overhauling of boiler 100% APH baskets should be taken out from their position and cleaned with high pressure water jet.

5.5.2 After baskets are cleaned and put back in position, these baskets should be covered with template to avoid any foreign material or dust going into the baskets.

5.5.3 All gaps between baskets and diaphragm are to be closed for better heat recovery and lower erosion rate at the edges by providing Basket-by-pass seals.

5.5.4 If the spare baskets are to be stored at site for a long time, it is recommended to apply rust preventive oil as specified by OEM, once in 6 months.

5.5.5 Before installation, all new heating elements are to be cleaned thoroughly with high pressure water jet / steam.

5.5.6 APH Diaphragm plates to be inspected thoroughly during overhauling to avoid any damage in running unit.

5.5.7 Soot blower travel is to be set to ensure full coverage of the entire heating element. Actual steam blowing is to be ensured visually. Thermal drain valve be provided at soot blowing steam line. Modifications may be done to provide additional electrical heater for super-heated steam during start up.

5.5.8 Parameter of Blowing medium to be ensured. If required, sequence of blowing one at a time to maintain steam pressure.

5.5.9 Soot blower nozzles should be replaced during every overhaul.

5.5.10 Soot Blower lance thickness should be checked during OH/ opportunity and to be replaced if thickness is less than 50% of original thickness.

5.5.11 In APHs, all eroded water washing manifolds/fire-fighting nozzles/deluge pipes shall be replaced during overhaul.

5.5.12 Healthiness of isolating valves provided for APH washing and fire-fighting to be ensured and its servicing to be carried out during OH.

5.5.13 Proper evacuation of Ash from economizer and APH hopper to be ensured during running units.

5.5.14 No bracing/protection angle/loose material is present in Eco to APH Flue gas path.

5.5.15 Before final box-up APH hopper to be cleaned.

Note: High DP across APH, high flue gas exit temperature, high air leakages must be discussed in daily planning/morning meeting along with financial loss.

5.6 BASKET REPLACEMENT CRITERIA

5.6.1 All the APH baskets should be taken out from their position and cleaned with high pressure water jet. After cleaning is over, the decision to replace APH baskets can be taken on the basis of the following visual inspection:

a. Erosion from top

b. Choking

c. Perforation in the elements

Each station must have rolling plan for replacement of Air preheater baskets in set as per following frequency.

d. Outermost two rows of Hot End baskets: Once in every 4 years.

e. 100% Baskets (Hot End, Hot Intermediate End & Cold End): Once in Every 8 years.

f. If condition of baskets are found more eroded or chocked prior to the replacement cycle, then following criteria to be considered:

- ✓ After the baskets are taken out from their position and cleaned, if the variation in weight is 20% and more, then the baskets should be replaced.
- ✓ If the weight of the baskets is found to be more than original weight, after high pressure jet cleaning, then baskets should be replaced.

5.6.2 Gas side efficiency, air side efficiency measurement using recommended procedure with grid sampling should be calculated before and after overhauling. Trend of these efficiencies may supplement the decision for basket replacement.

A. **Gas side efficiency for zero leakage**

$$\frac{\text{Gas temp. in - Gas Temp. Outlet at NL (No Leakage)}}{\text{Gas temp. in – Air temp. in}}$$

B. **Air side efficiency =** $\dfrac{\text{Air temperature rise}}{\text{Gas temp. in – Air temp. in}}$

C. **Air Heater X- Ratio** =

$$\frac{\text{Gas temp. in - Gas Temp. Outlet at NL (No Leakage)}}{\text{Air temp. out – Air temp. in}}$$

5.7 MONITORING OF APH PERFORMANCE

5.7.1 All seals to be replaced during every overhaul.

5.7.2 Necessary testing facilities for conductance of APH performance test (provision of ports, approach platform etc) should be created/confirmed jointly with EEMG.

5.7.3 BMD should be associated during the conductance of APH performance test.

5.7.4 Healthiness of on-line instruments Oxygen, FG/air temperature at APH inlet/outlet, pressure drop across APH should be ensured.

5.7.5 Reading from on-line instruments should be used only to monitor the trend but not for assessing degradation in performance. These readings should be validated during APH performance testing with off-line instruments.

5.8 MAINTENANCE OF DRIVE -SYSTEM

5.8.1 Visual inspection of gears and pinions by opening of top cover and side windows in everyoverhauling

5.8.2 All oil seals and 'O' rings are to be replaced during capital overhauling (for BHEL supplied gearboxes). For other gearboxes it should be done as per experience or OEM recommendations.

5.8.3 Replacement of Gear box internals (gear/pinion) to be carried out in set.

5.8.4 N.A

5.8.5 Fitment of input shaft & bearing to be checked for looseness & backlesh.

5.8.6 Oil to be replaced in every overhaul.

5.8.7 Inspection of input shaft keyway, clutch bearings, bibby coupling and Oil supply tube as per gear box model should be done in every overhaul and replaced as per defects.

5.8.8 Inspection of fluid coupling for any oil leakage however minute may be. During every overhaul, diaphragm seals, bearings are to be replaced. Special purpose test bed may be fabricated for foolproof performance of fluid coupling.

5.8.9 There is an integrated oil pump for lubrication of gear box top mounted bearing which is to be checked during each OH wherever it is available, its functions to be checked even during running.

5.8.10 Oil flow transmitters at bearing inlet line & alarm in control room to be provided.

5.8.11 Drive input shaft and its pinion to be procured and replaced in pair.

5.8.12 During overhauling Over-running clutch must be checked for bearing condition and replaced if required.

5.8.13 Inspection of pin rack for run out and excessive wear. Replacement of pins or pin rack to be planned in case of wear of case-hardened layer.

5.8.14 Welding of pin rack to Rotor shell to be checked, its DP to be carried out during every overhaul.

5.8.15 Pin gape of Last and first Pin to be checked at the adjacent Pin Rack (where segments joints) during pin rack replacement.

5.8.16 Pinion root gap, top and bottom gap are to be maintained as per OEM guidelines.

5.8.17 Inspection of main drive pinion for leveling and wear. Replacement of pinion to be planned in case of wear in the case-hardened layer.

5.8.18 Healthiness of taper lock bush and its bolts to be checked during every overhaul.

5.8.19 Taper lock bush bolts to be tightened through Torque wrench, with specified torque.

5.8.20 Ensure cold air flushing/sealing at rack pinion mating surface.

5.8.21 No lubricant is to be applied on pins or pinion.

5.8.22 Inspection / repair of air motor (liner/blade).

5.8.23 Alignment of electric motor, air motor & fluid coupling with gearbox to be carried out within limits.

5.8.24 Quarterly monitoring of gear box oil to be carried out for WDA.

Note: Health Matrix in enclosed format to be maintained and to be shared with COS.

6.0 PREVENTIVE MEASURES TO AVOID APH FIRE

6.1 Maintain proper atomizing medium pressure and oil pressure at the time of light up.

6.2 Maintain the required minimum oil temperature (>100 deg. C for HFO and LSHS). During light up, check for proper combustion by observing the flame and exit of the stack. Take corrective action in case of unstable flame.

6.3 In case of poor combustion of a particular burner, the reason should be identified and corrected immediately. Burners, which are not proving should not be tried repeatedly.

6.4 Air heater soot blowing should be started immediately after boiler light up. Blowing should be done with steam at proper pressure and temperature continuously for each air heater (including the idle air heater). During light up, if availability of soot blower is lost and cannot be restored back, boiler should be boxed up.

6.5 Oil carry over probes should be inspected regularly during light up, Free advance and retract of the probe to be checked during maintenance.

6.6 Cooling water line of oil carry over probe to be maintained healthy.

6.7 Availability of fire hydrants near air heater must be ensured.

6.8 Guide bearing oil level should not be overfilled as it may cause oil spill in guide bearing area and may cause fire. Oil level to be checked with LOP stopped condition. Mechanical maintenance should ensure proper marking on the dipstick.

6.9 Infrared fire detection system or any alternative system, for detecting fire at an early stage, should be available when air heater is in service, particularly during start up activities.

6.10 Thoroughness of fire fighting spray nozzles and coverage of all heating elements inside air heater must be checked before drying up of APH.

6.11 Operation Directive - COS-ISO-00- OD/OPS/SYST/004: Prevention of air Pre Heater Fires, is to be followed for preventing Fire in Air Preheaters.

7.0 BEST OPERATION AND COMMISSIONING PRACTICES FOR RELIABLE AND LONGER LIFE OF AIR PREHEATERS

7.1 Before Overhauling

7.1.1 Before starting of water washing, all the duct drains are to be opened. APH hoppers are to be emptied properly.

7.1.2 Water washing of air heaters should be carried out in hot condition after boiler shut down preferably when flue gas temperature is in between 200 to 150 deg C with draft fans stopped condition. Coverage, of water washing of all the heating elements is to be ensured before completion of water washing. pH value may be measured to determine effective cleaning. In the moment of equal pH value of inlet and outlet water, washing may be deemed to be complete.

7.1.3 Basket drying to be ensured by first using pressurized air.

7.2 During Re-commissioning

7.2.1 Before light up, cleanliness of the air heater and removal of all foreign materials should be ensured. Necessary protocols in this effect may be signed jointly by boiler maintenance and operation department.

7.2.2 Air heater dampers should be commissioned and made available for operation.

7.2.3 Air heater oil carry over probes and inspection ports with lights should be made available for operation.

7.2.4 Air heater fire and water washing lines should be charged up to the last valve.

7.2.5 Soot blower travel is to be checked jointly to ensure full coverage of all the heating elements. Actual steam blowing is to be ensured visually. Limit switch function to be checked.

7.2.6 All oil burners should be always checked for passing and defective ones must be rectified.

7.2.7 Infrared fire detection system or any alternative system, for detecting fire at an early stage, should be available when air heater is in service, particularly during startup activities.

7.2.8 Air heater interlocks and the functioning of the rotor stoppage alarm should be checked before the air heater is put in service.

7.2.9 Reference QP of APH overhauling works to be followed is enclosed herewith.

7.3 During Start Up

7.3.1 Boiler cold startup checklist covering all aspect as mentioned above should be signed with the concerned maintenance departments, before boiler light up.

7.3.2 Maintain proper atomizing medium pressure and oil pressure at the time of light up.

7.3.3 Maintain the required minimum oil temp (> 100 ^{0}C for HFO and LSHS). During light up check for proper combustion by observing the flame and exit of the stack. Take corrective action in case of unstable flame.

7.3.4 Air heater soot blowing should be started immediately after boiler light up. Blowing should be done with steam at proper pressure and temperature (8-10 ksc & 150 degree superheat) continuously for each air heater (including the idle air heater). During light up, if availability of soot blower is lost and cannot be restored, boiler should be boxed up.

7.3.5 Oil carryover probes should be inspected regularly during light up. (Oil carryover probes should be withdrawn and cooling water isolated once oil firing is stopped).

7.3.6 While the boiler and air heaters are in service, all the instruments related to combustion and air heaters should be monitored carefully and any sign of error in the instrument should be attended immediately.

7.3.7 A close watch should be kept on the terminal temperatures of air heaters. Any unusual increase in one or more of these should be investigated immediately.

7.4 During Normal Operation

7.4.1 Air motor should be run once in a week for thirty minutes to ensure healthiness. Ensure 6-7 drops of lub oil carried along with compressed air.

7.4.2 Guide bearing oil level should not be overfilled as it may cause oil spill in guide bearing area and may cause fire. This is to be checked with LOP in stopped condition. Mechanical maintenance should ensure proper marking on the dipstick.

7.4.3 Guide bearing area should be cleaned on daily basis.

7.4.4 Any abnormal sound from rotor, gear boxes or bearings should be intimated to mechanical maintenance immediately.

7.4.5 Soot blowing of air heater should be done at least once in a day with proper super-heated steam and with thermal drain valve in service. Frequency may be increased to thrice in a day (once in a shift) if APH DPs more than 50% than design.

7.4.6 At a time, soot blowers on only one side should be operated to ensure effective blowing at rated pressure. Opening of poppet valve on temperature interlock to be commissioned to ensure superheated steam quality during soot blowing operation.

7.4.7 All efforts must be made to maintain primary air header pressure close to OEM recommendation.

7.4.8 Drains after water washing and fire fighting lines after the isolating valves are to be kept opened for any accidental in rush of water inside the air pre-heater.

8.0 DEALING WITH AIR HEATER FIRE

Any sign of an air heater fire should be dealt with very seriously. On receiving a suspicion of air pre heater fire, start air heater soot blowers immediately (if not in progress already) and inspect the air heater locally for any abnormalities. If there is no sign of improvement and fire is confirmed, then proceed as follows:

8.1 Trip the boiler and draft plant and close all the associated dampers, leaving the air heater running.

8.2 Intimate fire tender.

8.3 Open fire water and water washing valves for extinguishing the fire.

8.4 Admit fire water in the air side also, through the manhole.

8.5 Continue the spray water until the rotor is completely cooled.

8.6 Keep the air heater running and open the inspection door when safe to examine the air heater.

8.7 Inspect the other air heater also.

NOTE:

- While admitting fire water, bottom duct/hopper drain / manhole should be opened. Damper also can be slightly opened for the escape of water and steam.
- The same procedure mentioned above is to be adopted, if the air heater is not rotating also

(jammed)

- Soot blowing operation can be suspended during fire fighting.
- Copious amount of water only and not other extinguishers like CO_2, halon etc. is effective during air-heater fire.

9.0 INSPECTION/CHECKS IN BOILER DURING OPPORTUNITIES/RSD

9.1 Operation of dampers may be checked at the time of unit shut down. APH gas inlet dampers to be operated to remove accumulated ash, if any.

9.2 Internal inspection of APH to be carried out, any loose bracings to be removed.

9.3 Cleaning of all filters of lub oil systems to be carried out.

9.4 Radial Seal, Bypass seal settings at HE to be carried out if required.

9.5 Inspection of Guide bearing, Support bearing or gearbox if PQ Index is beyond permissible limits.

NOTE:

1. All outages in Boiler auxiliaries like APH, irrespective of whether the outages result in partial loss or not, should be reported in enclosed C2 Format. This will enable

1.a) Understanding of damage mechanism

1.b) Establishment of root cause

1.c) Formulation of action plan to prevent similar failures in future.

1.d) Dissemination of earnings among all stations of NTPC.

2. Stations must communicate to OEM/OES if any equipment failure occurred during guarantee period with a copy to COS boiler and PE-Mech.

10.0 SPECIAL T&P FOR AIR HEATER MAINTENANCE

10.1 Gang jack (set of 4, Capacity - as per the rotor weight) with hydraulic pump, distributor manifold and all hoses in good working condition and 2 nos. spare jacks in working condition should be also readily available to cater to any failure of the working jacks in use. Hose should be minimum 3 meters long from distributor.

10.2 Master level (min 2 nos.) of accuracy, 0.02 or 0.0lmm / m.

10.3 Torque wrench of capacity 100kgf-m for 200 mw units and 200kgf-m for 500 MW units. (Min 2 nos. per site.)

10.4 Fork lift (min. 2 nos. of capacity 3.0 T approx.) to handle baskets at zero meter.

10.5 Electric hoists for handling basket replacement / cleaning works. One hoist for each Air Heater for removing / installing baskets in the Air Heater plus one common hoist for transporting baskets to & fro from zero meter.

10.6 Pneumatic wrenches (min. 8 sets) for fasteners, for seal replacement & cold end basket doors removal / installation works.

10.7 Aluminum straight edges for seal setting of radial and axial seals. (Min 4 nos. for radial and 2 no. for axial)

10.8 Two nos. of electrical winch machines per site.

10.9 Heating coil/Induction coil one number.

10.10 Hydraulic ejector/pump one number.

10.11 Space to be identified and earmarked exclusively at zero-meter for handling, repair and transportation of baskets.

11.0 SAFETY

11.1 SAFETY HAZARDS AND PRECAUTIONS TO BE TAKEN:

All personnel inspecting Air Preheater components should be aware of the following safety hazards so as to prevent injury to personnel:

- Tripping hazard when entering / exiting the rotor assembly
- Confined space area
- Loose/broken walkways / ledges
- Slipping hazards
- Falling debris hazards
- Hazards from sharp corners/ eroded components
- Use of wooden planks as scaffolding material should be discontinued.
- Mandatory use of safety belts by people working at heights.
- Erection of safety nets below ducts/openings.
- Barricading of all openings
- Load testing of lifting tackles, supporting structure by safety/FQA/Third party before usage in OHs

11.2 SAFETY CHECKLIST FOR APH INSPECTION / MAINTENANCE:

- All personnel to have proper PPE (Safety shoes, safety helmet, hand gloves, ear plugs, nose masks, boiler suit etc.,)
- The area being inspected is properly lit.
- Personnel working at heights should wear safety belt.
- Proper scaffolding or walkways / planking are in place and secure to stand / walk on.
- Erection of scaffolding covering 100% area on Cold end.
- Safety nets should be provided while working at heights.
- All ropes, chains, cables, slings and other hoist equipment must be inspected each time before use.

- Only qualified personnel should be assigned to rigging operation.
- Floor openings should be guarded by a standard rail and toe board or cover.
- Wall openings should have a standard guard rail.
- To ensure 100% compliance of safety and quality system, each executive in boiler including senior executive must ensure minimum 02hrs physical presence on daily basis at critical Boiler & Auxiliary locations during overhaul and any forced outage.
- For round the clock monitoring CCTV cameras to be installed at critical locations during overhaul.

12.0 REVIEW

The Head of Project will be responsible for reviewing this document on a 2-yearly basis or as necessary.

13.0 Quality Plan

NTPC	NAME OF WORK: AIR-PREHEATER OVERHAULING / MAINTENANCE WORKS	STANDARD QUALITY CHECKS Confirming to: COS-ISO-00-OGN-OPS-SYST-018 Rev.00 COS-ISO-00-OIN/OPS/MECH/023 Rev. No. : 0, OS/BLR/Rotory Parts/ 2016 dated 12/02/2016	QP No. OS/BOILER/APH/001 Rev. 00 Date: Jan'2018 Page 1 of 2	Prepared by: Reviewed by:	Approved by:

Sl. No.	Component / Operation	Type of Check	Class	Quantum of Check	Ref. Document / Acceptance Norm	Format of Record	Agency				Remarks
							A	B	C	O	
1.0	Check & Clean Baskets by HP water jet(Hot End, Hot Intermediate, Cold End)	Visual / Lamp	Major	100%	O & M Manual / Drg.	Log Sheet		W	P	W	
2.0	Check For Erosion / damage / Welding	Visual /	Major	100%				W	P		
2.1	Protective water shield at support bearing & Guide Bearing	Visual	Major	100%	O & M Manual / Drg.	Log Sheet		W	P		
2.2	Baskets Frame, Support grills (gratings), Expansion Cover plate, Support structure, Rotor diaphragm, Duct Bracing and Damper flap	Visual	Major	100%	O & M Manual / Drg.	Log Sheet		W	P		
2.3	Mating Duct Plate & Thickness	Visual / D-metre	Major	100% 60%	O & M Manual / Drg.	Log Sheet	V	W	P		Thickness to be checked for erosion prone area.
2.4	Expansion Bellow Before & After APH	Visual / ATT	Major	100%	O & M Manual / Drg.	Log Sheet		W	P	W	ATT to be carried out after completion of work.
2.5	Fire water, APH wash, Soot Blowing Nozzles and its Manifold , Oil Carryover probe	Visual	Major	100%	O & M Manual / Drg.	Log Sheet		W	P		
2.6	All seals including Static seals.	Visual	Major	100%	O & M Manual / Drg.	Log Sheet		W	P		
3.0	Inspection, Servicing And Clearance Measurement										
3.1	Guide bearing and support bearing (Measurement of critical dimension as per manual)	Visual,	Major	100%	O & M Manual / Drg.	Log Sheet	W	W	P		Blue matching of adapter sleeve with trunnion shaft on replacement of Guide Bearing
3.2	Drive inspection & alignment	Visual, Measure	Major	100%	O & M Manual / Drg.	Log Sheet	W	W	P		

एनटीपीसी NTPC	NAME OF WORK: AIR-PREHEATER OVERHAULING / MAINTENANCE WORKS	STANDARD QUALITY CHECKS Confirming to: COS-ISO-00-OGN-OPS-SYST-018 Rev.00 COS-ISO-00-OIN/OPS/MECH/023 Rev. No. : 0, OS/BLR/Rotory Parts/ 2016 dated 12/02/2016	QP No. OS/BOILER/APH/001 Rev. 00 Date: Jan'2018 Page 2 of 2	Prepared by: Reviewed by:	Approved by:

Sl. No	Component / Operation	Type of Check	Class	Quantum of Check	Ref. Document / Acceptance Norm	Format of Record	Agency				Remarks
							A	B	C	O	
3.3	Lubricant Wear debris test	WDA	Major	100%	Quarterly/Annum as per OS guide line	Log Sheet	W	W			To be done at NETRA Pre OH is also required
3.4	Rotor levelling	Measure	Major	100%	O & M Manual / Drg.	Log Sheet	W	W	P		Rotor leveling within 0.1 mm / mtr or as OEM
3.5	Sector Plate levelling	Measure	Major	100%	O & M Manual / Drg.	Log Sheet	W	W	P		within +/- 0.5mm
3.6	Seal clearances	Measure	Major	100%	O & M Manual / Drg.	Log Sheet	V	W	P		
3.7	Free Thermal Expansion *	Visual, Measure	Major	100%	O & M Manual / Drg.	Log Sheet		W	P		Expansion study with help of aluminium finger tips
3.8	Soot blower Lance & Nozzle, Oil Carryover probe	Visual, Measure	Major	100%	O & M Manual / Drg.	Log Sheet		W	P	W	

LEGEND:

A – FQA & EIC, B – Execution department, O – Operation department, C – Contractor, P – Perform, W – Witness, V – Verification

Note: 1. Checks shall be applicable as per the scope of work identified by EIC.
2. It is the responsibility of the contracting agency to offer 'A' class checks to witness by FQA and EIC & 'B' class checks to witness by Execution department and 'Ó' class by Operation department. The complete protocols are to be made on real time basis.
3. Welding electrodes shall be of NTPC approved make and brand only.
* To ensure Free Thermal Expansion, after fixing all seals sector plates are to be lifted for 10-15mm to check the freeness of guide bearing Spool and afterwards proper levelling to be carryout.

14.0 AIR PREHEATER HEALTH MATRIX

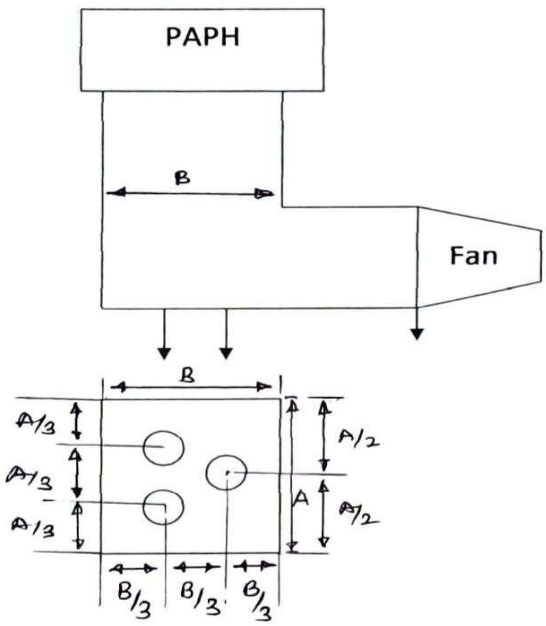
APHs Health Assessment and Replacement Monitoring Matrix												
Station												
Equipment's	Date of last brg replacement		Last basket replacement date (100%)	Last Date of inspection/ defects	WDA report of (GB)		WDA report of (SB)		WDA report of (Gear Box)		Bearing Temperature	
	Guide	Support			PQ	Date	PQ	Date	PQ	Date	GB	SB

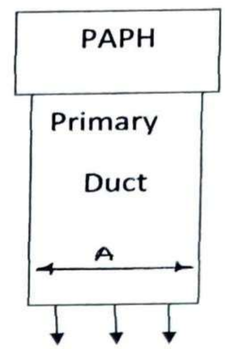
15.0 AVAILABILTY OF CRITICAL SPARES

Availability of Critical spares as per COS guidelines	
Spare's	**Stock**
Guide bearing	
Support bearing	
Gear Box	
Seals	

Annexure-I

Unit size (500/660/800)MW





1. Diameter of the pipe-200mm.
2. Length of the pipe-300mm.
3. Dummy flange to be used with bolting.
4. Between flanges a gasket to be used.